Assignment No 1

Subject :- Agentic AI

Roll No :- B-26

Aim :- Develop a Python program that simulates an AI agent performing tasks. Store each state transition using a dictionary inside a list.
Requirements

- Agent ID
- Current State
- Previous State
- Timestamp
- Action Performed

Code :-

```
from datetime import datetime
agent_id = "AI001"
states = [
   ("Idle", "Agent Started"),
   ("Collecting Data", "Collecting Input Data"),
   ("Processing", "Processing Data"),
   ("Decision Made", "Decision Generated"),
   ("Completed", "Task Completed")
]
agent_history = []
previous_state = "None"

for state, action in states:
```

```
    transition = {
        "Agent ID": agent_id,
        "Current State": state,
        "Previous State": previous_state,
        "Timestamp": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        "Action Performed": action
    }

    agent_history.append(transition)
    previous_state = state

print("Agent State History\n")

for i, record in enumerate(agent_history, start=1):
    print(f"Step {i}")
    print(f"State : {record['Current State']}\n")

print("=" * 50)
print("Complete Agent History\n")

for i, record in enumerate(agent_history, start=1):
    print(f"Step {i}")
    print(f"Agent ID        : {record['Agent ID']}")
    print(f"Current State    : {record['Current State']}")
    print(f"Previous State   : {record['Previous State']}")
    print(f"Timestamp       : {record['Timestamp']}")
    print(f"Action Performed : {record['Action Performed']}")
    print("-" * 50)
```

```
print("\nAgent History in Reverse Order\n")
for i, record in enumerate(reversed(agent_history), start=1):
    print(f"Step {i}")
    print(f"Agent ID        : {record['Agent ID']}")
    print(f"Current State    : {record['Current State']}")
    print(f"Previous State   : {record['Previous State']}")
    print(f"Timestamp       : {record['Timestamp']}")
    print(f"Action Performed : {record['Action Performed']}")
    print("-" * 50)
```

Output :-

```
Agent State History

Step 1
State : Idle

Step 2
State : Collecting Data

Step 3
State : Processing

Step 4
State : Decision Made

Step 5
State : Completed

================================================
Complete Agent History

Step 1
Agent ID         : AI001
Current State    : Idle
Previous State   : None
Timestamp        : 2026-07-16 14:39:55
Action Performed : Agent Started
------------------------------------------------
Step 2
Agent ID         : AI001
Current State    : Collecting Data
Previous State   : Idle
Timestamp        : 2026-07-16 14:39:55
Action Performed : Collecting Input Data
------------------------------------------------
Step 3
Agent ID         : AI001
Current State    : Processing
Previous State   : Collecting Data
Timestamp        : 2026-07-16 14:39:55
Action Performed : Processing Data
------------------------------------------------
Step 4
Agent ID         : AI001
Current State    : Decision Made
Previous State   : Processing
Timestamp        : 2026-07-16 14:39:55
Action Performed : Decision Generated
------------------------------------------------
Step 5
Agent ID         : AI001
Current State    : Completed
Previous State   : Decision Made
Timestamp        : 2026-07-16 14:39:55
Action Performed : Task Completed
```

```
------------------------------------------------

Agent History in Reverse Order

Step 1
Agent ID         : AI001
Current State    : Completed
Previous State   : Decision Made
Timestamp        : 2026-07-16 14:39:55
Action Performed : Task Completed
------------------------------------------------
Step 2
Agent ID         : AI001
Current State    : Decision Made
Previous State   : Processing
Timestamp        : 2026-07-16 14:39:55
Action Performed : Decision Generated
------------------------------------------------
Step 3
Agent ID         : AI001
Current State    : Processing
Previous State   : Collecting Data
Timestamp        : 2026-07-16 14:39:55
Action Performed : Processing Data
------------------------------------------------
Step 4
Agent ID         : AI001
Current State    : Collecting Data
Previous State   : Idle
Timestamp        : 2026-07-16 14:39:55
Action Performed : Collecting Input Data
------------------------------------------------
Step 5
Agent ID         : AI001
Current State    : Idle
Previous State   : None
Timestamp        : 2026-07-16 14:39:55
Action Performed : Agent Started
------------------------------------------------
PS D:\Agentic AI Practical>
```